

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from pandas.core.common import random_state
from sklearn.linear_model import LinearRegression

df_sal = pd.read_csv('Salary_Data.csv')
df_sal.head()

```

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0

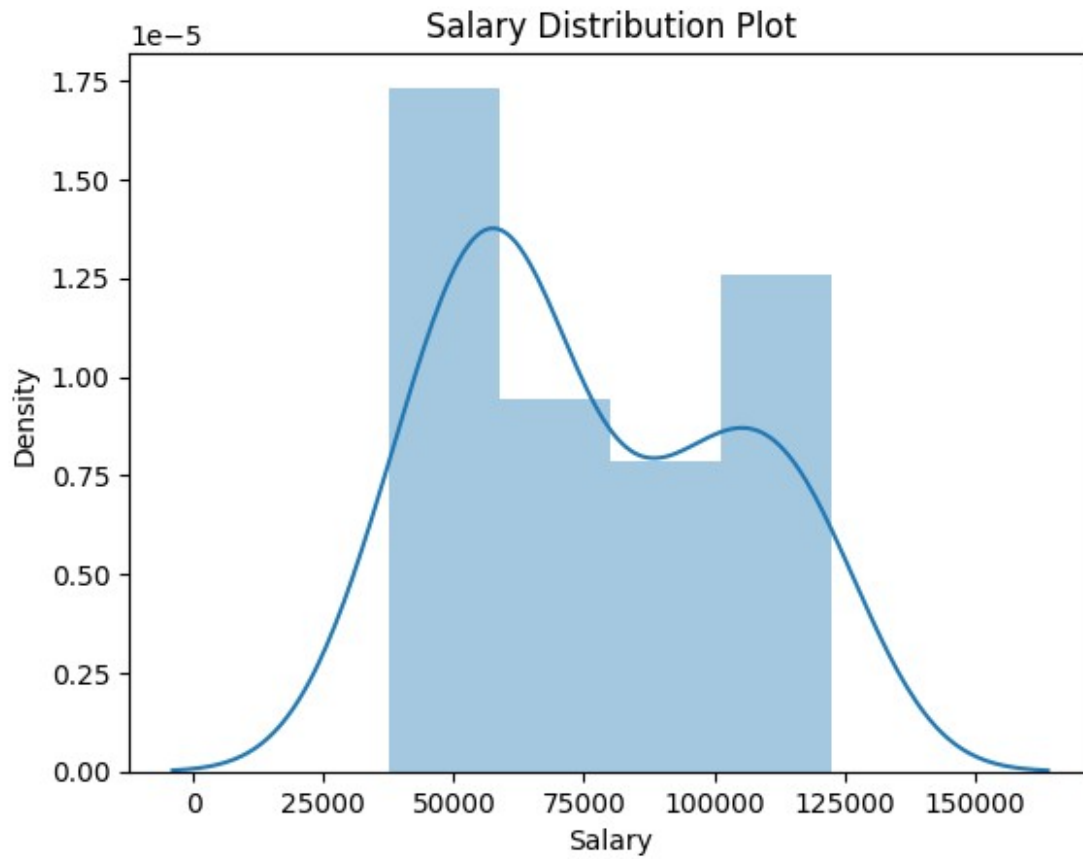
```
df_sal.describe()
```

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```

plt.title('Salary Distribution Plot')
sns.distplot(df_sal['Salary'])
plt.show()

```



```
plt.scatter(df_sal['YearsExperience'], df_sal['Salary'], color =  
'lightcoral')  
plt.title('Salary vs Experience')  
plt.xlabel('Years of Experience')  
plt.ylabel('Salary')  
plt.box(False)  
plt.show()
```



```
import pandas as pd

# Example: Create df_sal from a CSV file
df_sal = pd.read_csv("Salary_Data.csv") # replace with your actual
file path

# Now you can split the data
X = df_sal.iloc[:, :1] # independent variable(s)
y = df_sal.iloc[:, 1:] # dependent variable(s)

import pandas as pd
from sklearn.model_selection import train_test_split

# Sample DataFrame (replace this with your actual data loading method)
# Example data with one independent and one dependent variable
data = {
    'Experience': [1, 2, 3, 4, 5],
    'Salary': [30000, 35000, 40000, 45000, 50000]
}

df_sal = pd.DataFrame(data)

# Split features and target
```

```

X = df_sal.iloc[:, :1] # independent variable
y = df_sal.iloc[:, 1:] # dependent variable

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=0)

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Sample Data
data = {
    'Experience': [1, 2, 3, 4, 5],
    'Salary': [30000, 35000, 40000, 45000, 50000]
}
df_sal = pd.DataFrame(data)

# Feature and Target
X = df_sal.iloc[:, :1] # Experience
y = df_sal.iloc[:, 1:] # Salary

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=0)

# Fit Linear Regression Model
regressor = LinearRegression()
regressor.fit(X_train, y_train)

LinearRegression()

# Step 1: Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error

# Step 2: Load the dataset
dataset = pd.read_csv('Salary_Data.csv') # Ensure this file is in
your working directory
X = dataset.iloc[:, :-1].values # Independent variable
(Years of Experience)
y = dataset.iloc[:, -1].values # Dependent variable
(Salary)

# Step 3: Split dataset into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=0)

```

```

# Step 4: Train the Linear Regression model
regressor = LinearRegression()
regressor.fit(X_train, y_train)

# Step 5: Predict test set results
y_pred = regressor.predict(X_test)

# Step 6: Evaluate the model
mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)

print("==== Model Evaluation ====")
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"Mean Absolute Error (MAE): {mae:.2f}")
print(f"Coefficient (Slope): {regressor.coef_[0]:.2f}")
print(f"Intercept: {regressor.intercept_:.2f}")
print("=====")

==== Model Evaluation ====
Mean Squared Error (MSE): 12823412.30
Mean Absolute Error (MAE): 2446.17
Coefficient (Slope): 9312.58
Intercept: 26780.10
=====

# Step 7: Visualize Training Set Results
plt.figure(figsize=(8, 5))
plt.scatter(X_train, y_train, color='green', label='Training Data')
plt.plot(X_train, regressor.predict(X_train), color='blue',
label='Regression Line')
plt.title('Salary vs Experience (Training Set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.legend()
plt.grid(True)
plt.show()

```



Step 8: Visualize Test Set Results

```
plt.figure(figsize=(8, 5))
plt.scatter(X_test, y_test, color='red', label='Actual (Test Data)')
plt.plot(X_train, regressor.predict(X_train), color='blue',
label='Regression Line')
plt.title('Salary vs Experience (Test Set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.legend()
plt.grid(True)
plt.show()
```



```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
dataset = pd.read_csv('Salary_Data.csv')
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=0)
```

```
regressor = LinearRegression()
regressor.fit(X_train, y_train)
```

```
y_pred = regressor.predict(X_test)
#Root Mean Squared Error (RMSE):
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
print(f"Root Mean Squared Error (RMSE): {rmse}")
```

Root Mean Squared Error (RMSE): 3580.979237321343

```
#R2 Score:
r2 = r2_score(y_test, y_pred)
print(f"R2 Score: {r2}")
```

R² Score: 0.988169515729126